Flight Delays & Cancellations

Analysis Report

Part 1 — Data Cleaning

Before building the dashboard, the raw dataset was cleaned and prepared using Python. The steps below cover each stage of the process.

Step 1 — Import Libraries & Load Data

The required libraries (Pandas, NumPy, Matplotlib) are imported and the raw CSV file is loaded. A quick look at the shape and first rows helps understand the structure before any processing.

Step 2 — Check for Missing Values & Duplicates

Each column is scanned for missing values and the percentage of nulls is calculated. Duplicate rows are also identified so nothing is counted twice in the analysis.

Step 3 — Clean & Fill Missing Data

Missing values are handled column by column — some are filled with the median or mode, others are dropped if they are too sparse to be useful. Date and time columns are converted to the correct datetime format.

Step 4 — Feature Engineering

New columns are created from existing ones: hour of day, day of week, month, and season are extracted from the timestamp. A binary 'delayed' flag is added based on a defined threshold to make filtering easier in the dashboard.

Step 5 — Filter Outliers & Save Clean File

Rows with unrealistic values (e.g. negative delay times) are removed. Columns are renamed for clarity and the clean dataset is saved as a new CSV file ready for the dashboard.

Step 6 — Final Quality Check

A final check confirms there are no remaining nulls, all column types are correct, and the summary statistics look reasonable before moving on to the dashboard.

Figure 1: Importing libraries and loading the dataset

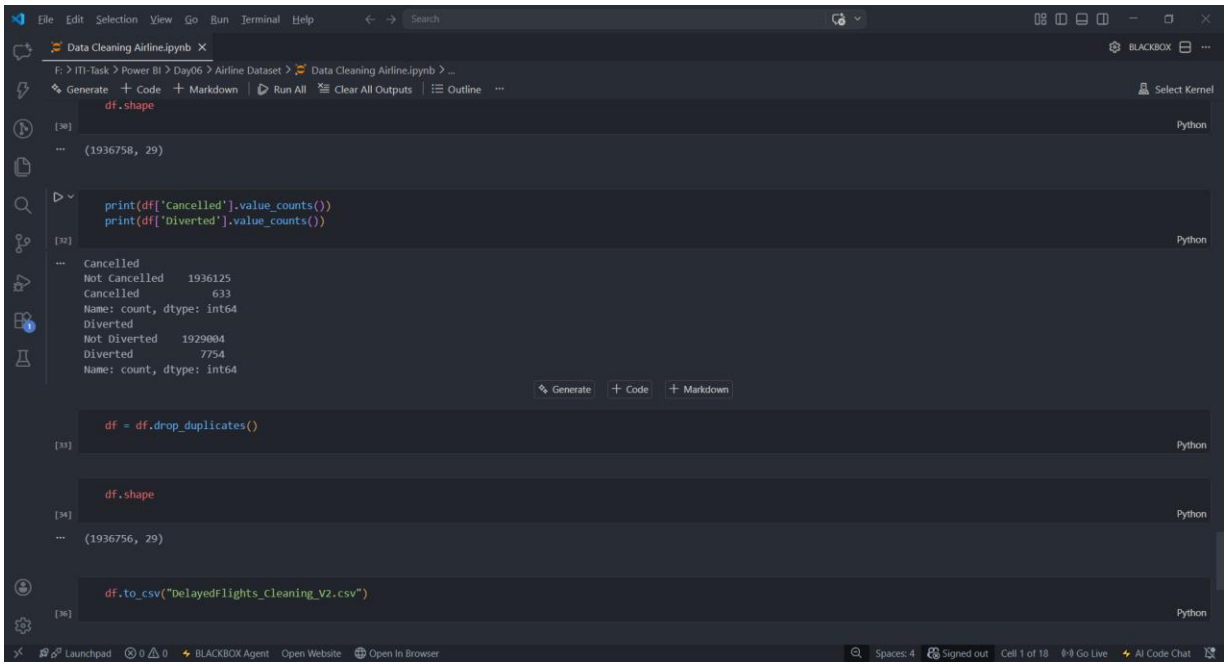


Figure 2: Missing values report and duplicate check

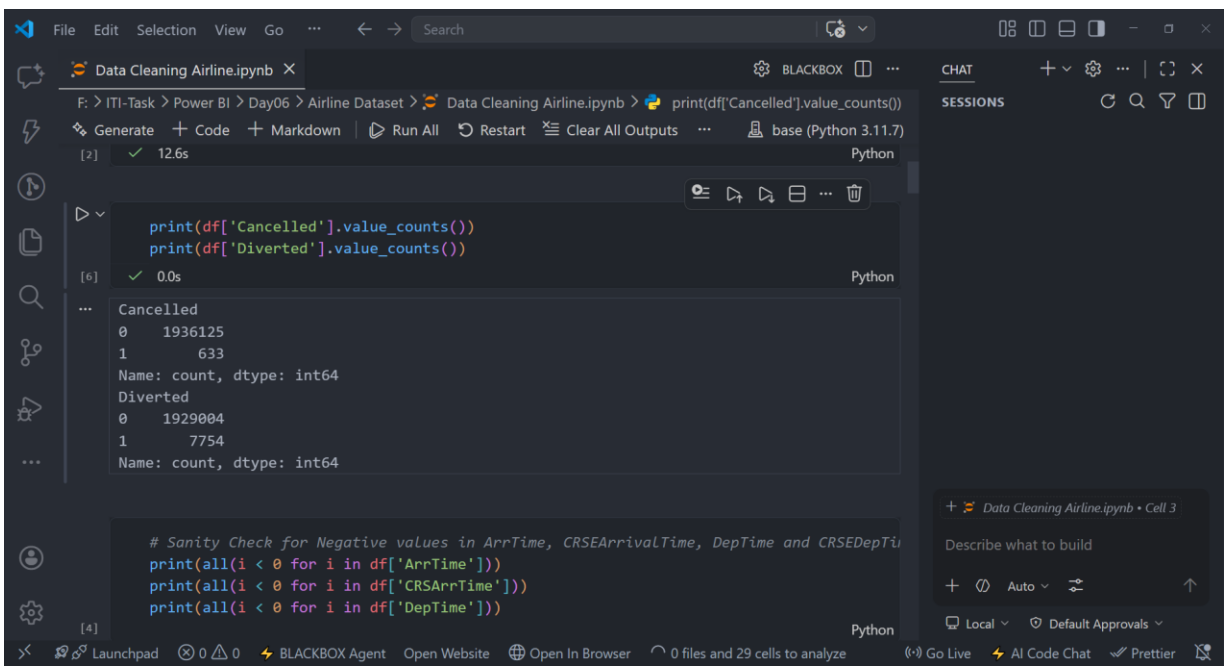


Figure 3: Handling missing values and fixing column types

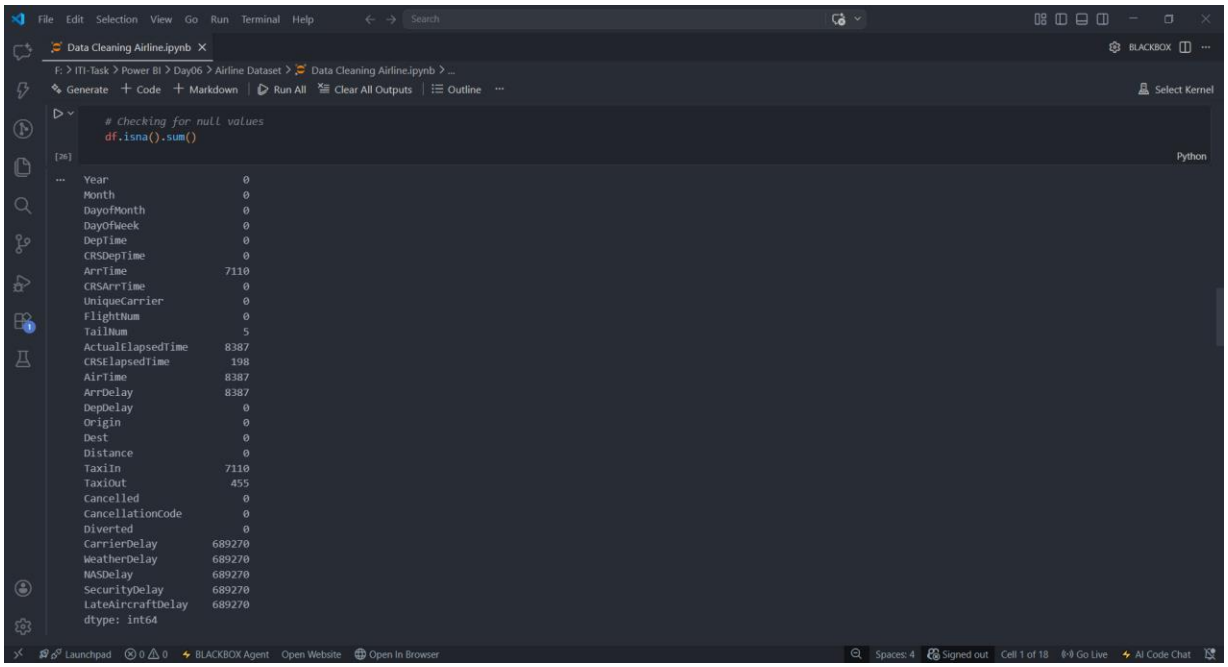


Figure 4: Extracting time-based features and creating the delay flag

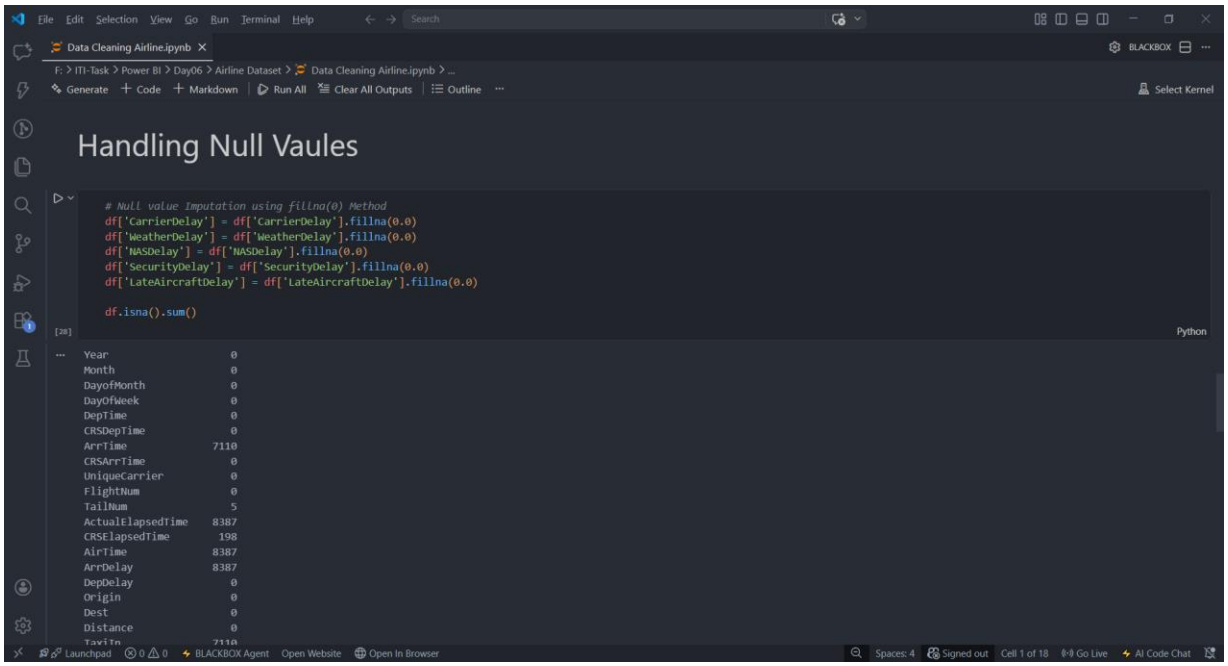


Figure 5: Removing outliers and exporting the clean dataset

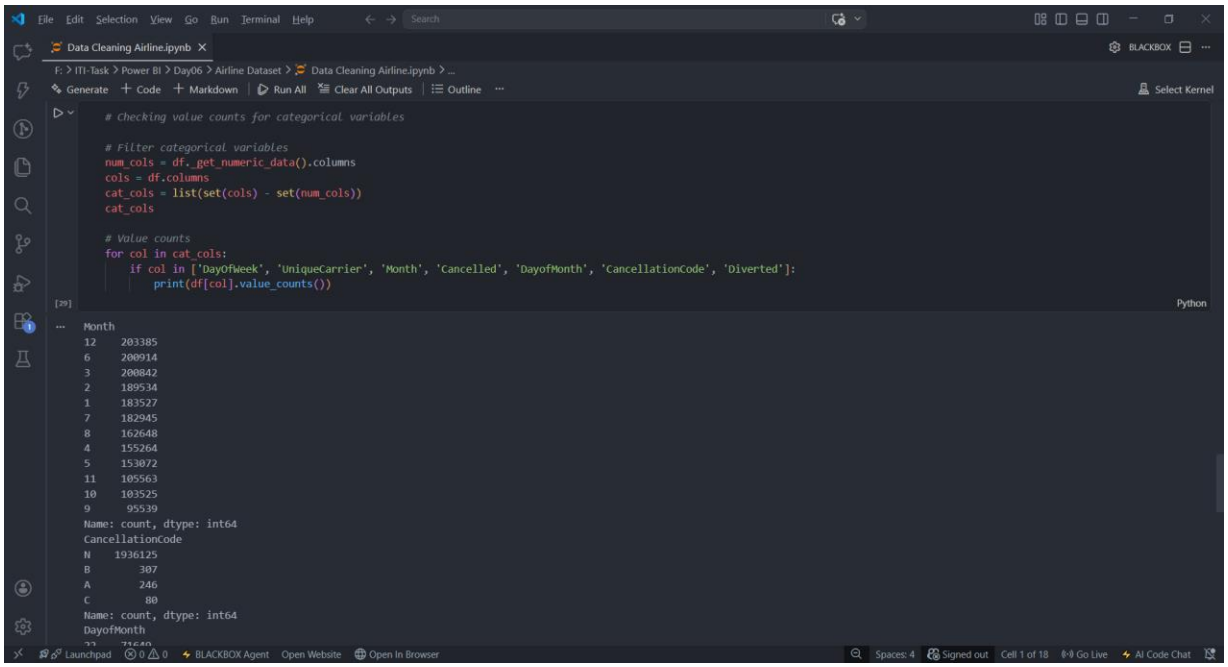


Figure 6: Final quality check and descriptive statistics

Part 2 — Dashboard

The clean dataset was used to build an interactive dashboard. Each page focuses on a different angle of the data.

Page 1 — Overview

The overview page shows the main KPIs at a glance: total flights, percentage delayed, percentage cancelled, and average delay in minutes. A high-level chart shows how flight statuses are distributed across the full time period.

Page 2 — Delay Reasons

This page breaks down delays by cause — carrier delays, late aircraft, NAS (air traffic), weather, and security. A bar chart ranks each cause by total minutes of delay so the biggest contributors are easy to spot.

Page 3 — Cancellations

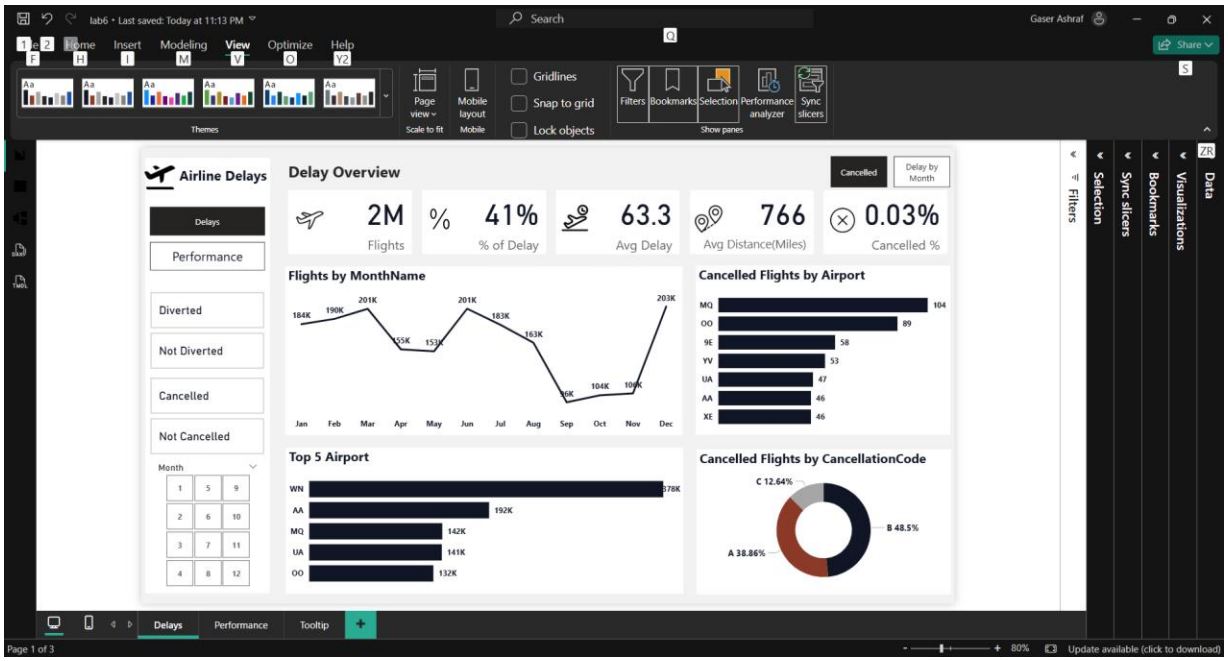
The cancellations page shows which airlines cancel the most flights, which months see the highest cancellation rates, and what the main cancellation codes are (weather, carrier, or NAS).

Page 4 — Time Trends

Line and bar charts show how delays and cancellations change over time — by month, day of week, and hour of day. Clear peaks are visible during summer months and early morning departures.

Page 5 — Airline Performance

The final page compares airlines side by side on delay rate, cancellation rate, and average delay time. The comparison is normalized by flight volume so larger airlines are not unfairly penalized.


Airline Delays
Delay Overview
Cancelled
Delay by Month
2M
Flights
41%
% of Delay
63.3
Avg Delay
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Avg Distance(Miles)
0.03%
Cancelled %
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Diverted
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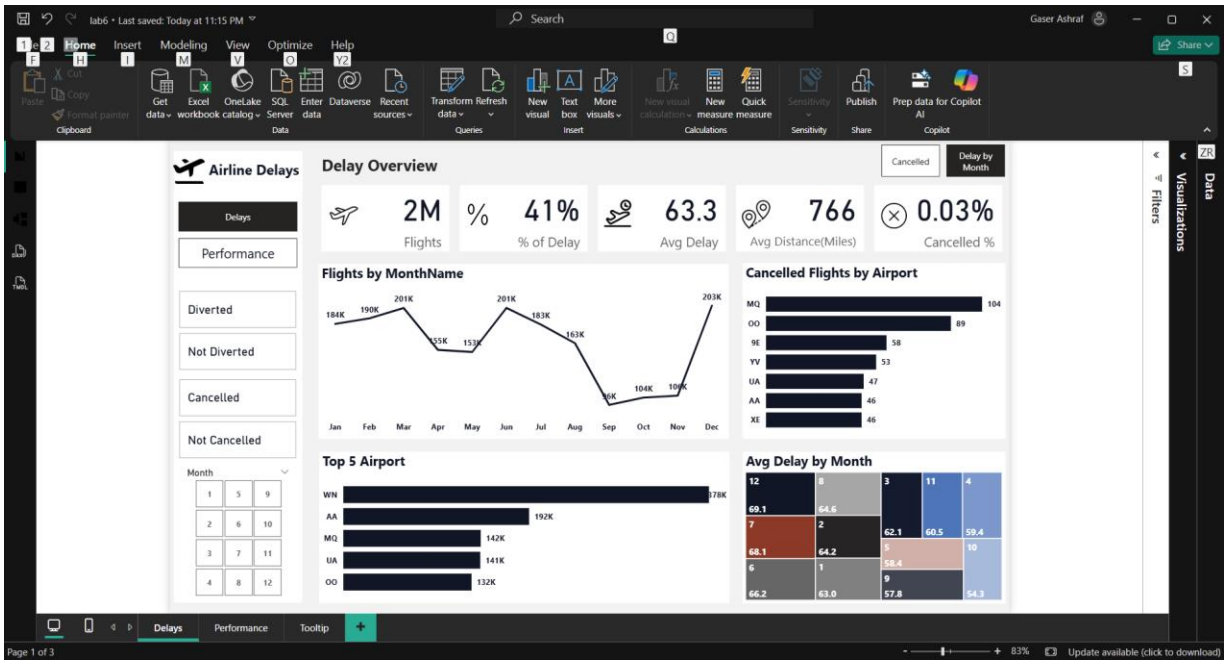
Cancelled
Not Cancelled
Month
Flights by MonthName
Cancelled Flights by Airport
Top 5 Airport
Cancelled Flights by CancellationCode
C 12.64%
B 48.5%
A 38.86%
Delays
Performance
Tooltip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Avg Delay by Month
Delays
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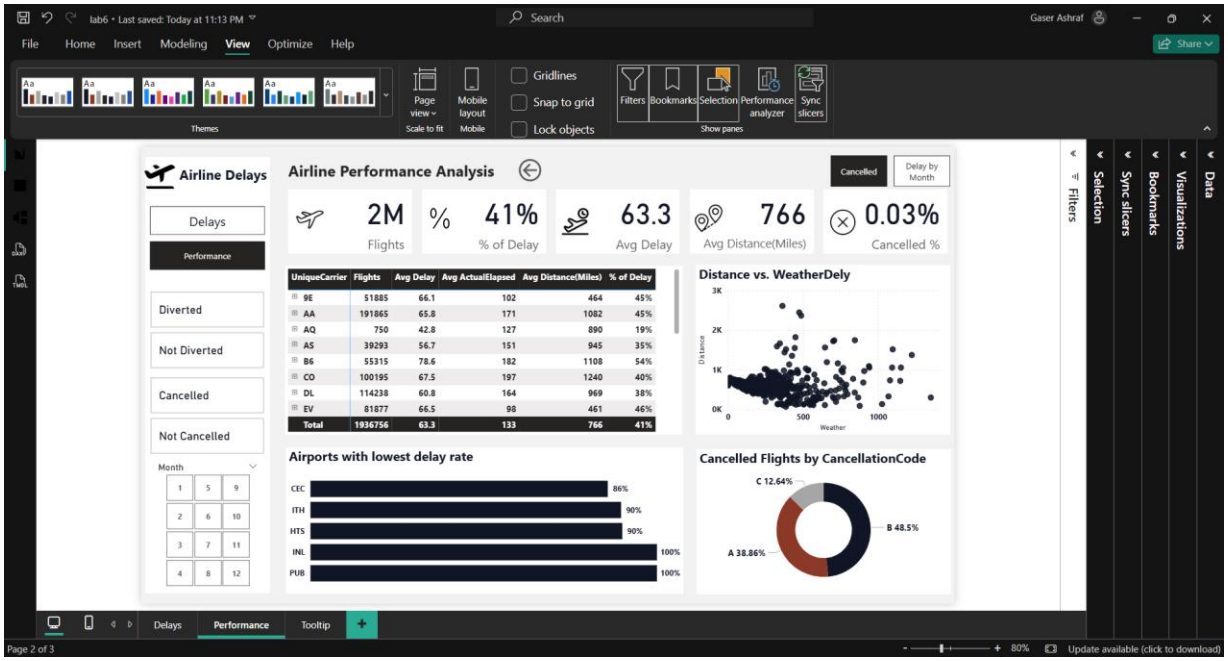
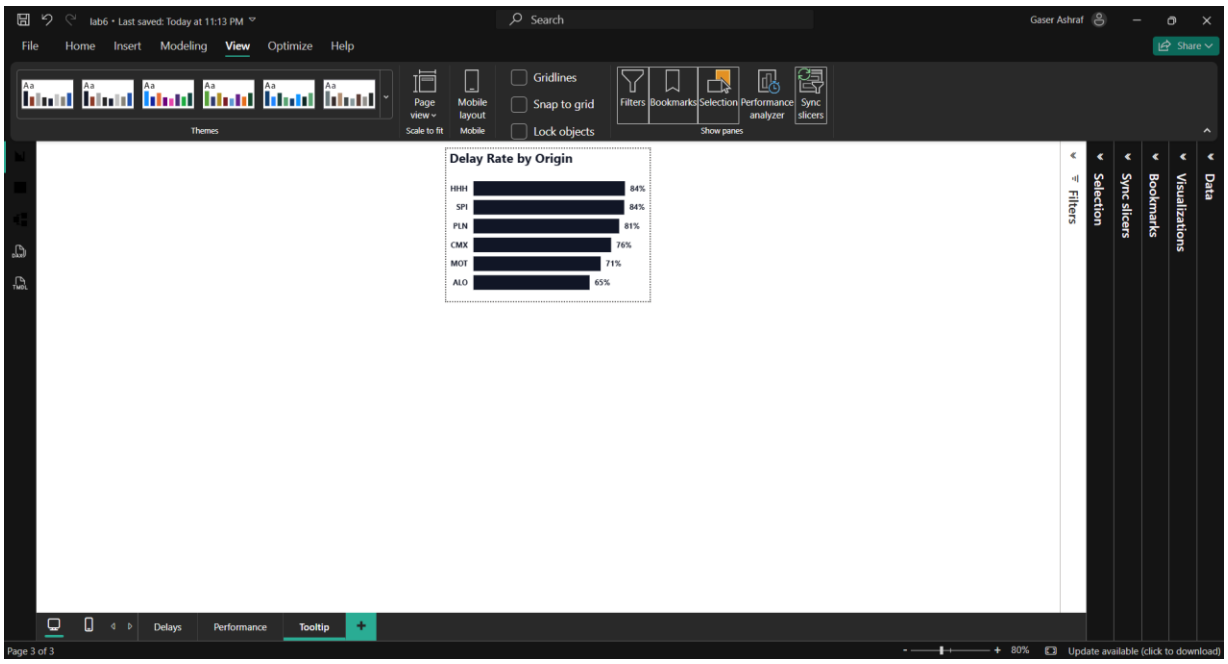
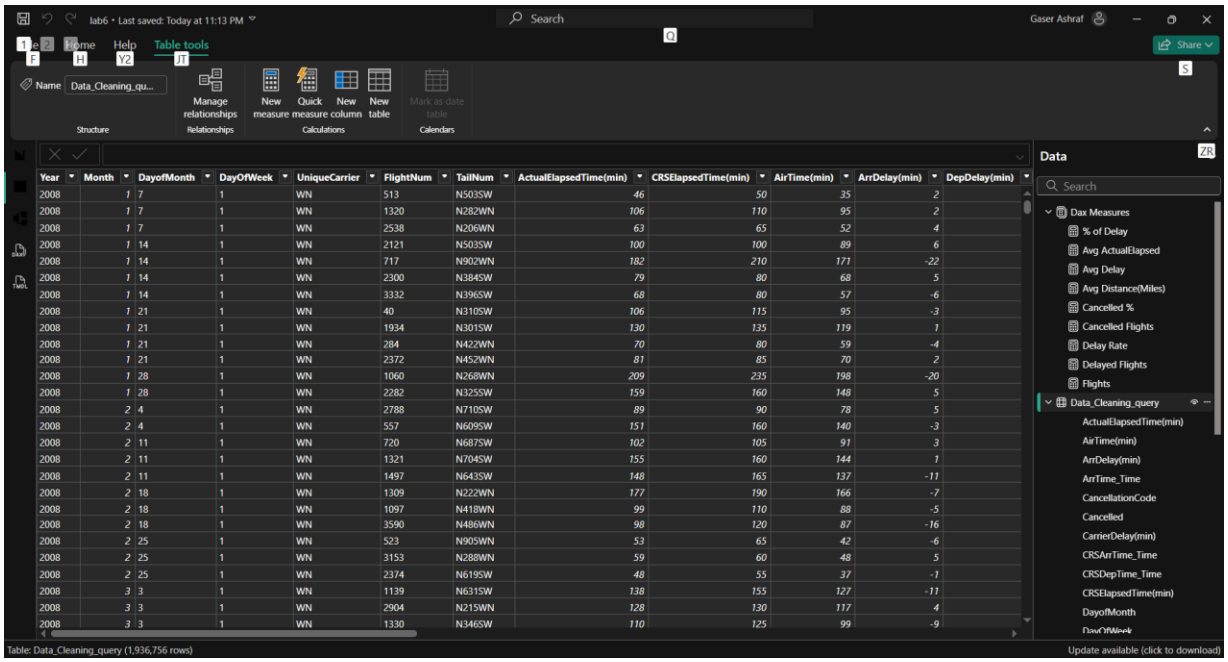
Tooltip

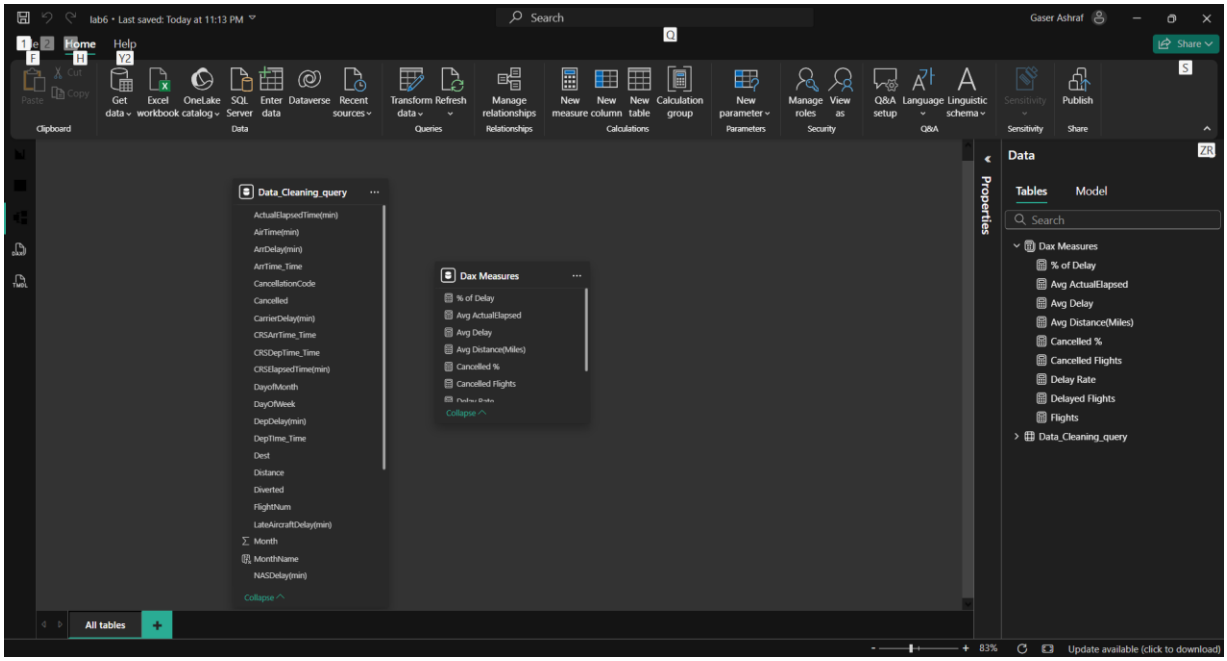
DrillThrow and Bookmark







Year	Month	DayofMonth	DayOfWeek	UniqueCarrier	FlightNum	TailNum	ActualElapsedTime(min)	CRSElapsedTime(min)	AirTime(min)	ArrDelay(min)	DepDelay(min)
2008	1	7	1	WN	513	N503SW	46	50	35	2	
2008	1	7	1	WN	1320	N282WN	106	110	95	2	
2008	1	7	1	WN	2538	N206WN	63	65	52	4	
2008	1	14	1	WN	2121	N503SW	100	100	89	6	
2008	1	14	1	WN	717	N902WN	182	210	171	-22	
2008	1	14	1	WN	2300	N384SW	79	80	68	5	
2008	1	14	1	WN	3332	N396SW	68	80	57	-6	
2008	1	21	1	WN	40	N310SW	106	115	95	-3	
2008	1	21	1	WN	1934	N301SW	130	135	119	1	
2008	1	21	1	WN	284	N422WN	70	80	59	-4	
2008	1	21	1	WN	2372	N452WN	81	85	70	2	
2008	1	28	1	WN	1060	N268WN	209	235	198	-20	
2008	1	28	1	WN	2282	N325SW	159	160	148	5	
2008	2	4	1	WN	2788	N710SW	89	90	78	5	
2008	2	4	1	WN	557	N609SW	151	160	140	-3	
2008	2	11	1	WN	720	N687SW	102	105	91	3	
2008	2	11	1	WN	1321	N704SW	155	160	144	1	
2008	2	11	1	WN	1497	N643SW	148	165	137	-11	
2008	2	18	1	WN	1309	N222WN	177	190	166	-7	
2008	2	18	1	WN	1097	N418WN	99	110	88	-5	
2008	2	18	1	WN	3590	N486WN	98	120	87	-16	
2008	2	25	1	WN	523	N905WN	53	65	42	-6	
2008	2	25	1	WN	3153	N288WN	59	60	48	5	
2008	2	25	1	WN	2374	N619SW	48	55	37	-1	
2008	3	3	1	WN	1139	N631SW	138	155	127	-11	
2008	3	3	1	WN	2904	N215WN	128	130	117	4	
2008	3	3	1	WN	1330	N346SW	110	125	99	-9	



Key Takeaways

- Late aircraft is the single biggest driver of delays.
- Summer months and morning peak hours see the highest delay rates.
- Weather delays are less frequent but cause the longest average delay time.
- There is a wide performance gap between airlines — improvement is possible.
- Thanks to clean and organized data, dashboard insights have become more reliable.